

Jacob Lamadrid

Machine Learning | Data Science | Systems Analytics | Research

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EDUCATION

•Master of Science in Analytics

Georgia Institute of Technology

Atlanta, GA

Expected: Dec 2027

- Focus: Modeling, Simulation, and Optimization

•Bachelor of Science in Cognitive Science - Machine Learning and Neural Computation

University of California, San Diego

San Diego, CA

Jun 2024

- Minor: Data Science
- Provost Honors | 2024 Cognitive Science Summer Scholar (\$6,000 Award)
- Instructional Assistant: Introduction to Python, Introduction to Data Science

EXPERIENCE

•Graduate Data Science Intern

Centene Corporation

May 2026 – Aug 2026

St. Louis, MO

- Developed end-to-end data engineering pipelines unifying Internal Audit and claims warehouse sources into a common schema of 127K+ claims for anomaly detection on large-scale health insurance claims data
- Independently designed and implemented an AutoEncoder-based weakly-supervised anomaly detection method in PyTorch, achieving an 10x increase in precision of identifying erroneous claims over the prior baseline on a 50:1 class-imbalanced dataset

•Research Fellow

UC San Diego Natural Computation Lab

Jul 2024 – Sep 2024

San Diego, CA

- Conducted independent research on audio reconstruction from EEG signals using CNNs, Latent Diffusion Models, and DCGANs; managed full ML lifecycle from multimodal data ingestion to model evaluation
- Designed signal processing pipelines for EEG and audio data; translated theoretical research into reproducible PyTorch experiments on custom neuroimaging datasets

•Application Developer & Research Data Associate

UC San Diego Health – Autism Center of Excellence

Jun 2022 – Sep 2024

San Diego, CA

- Built TrackiT!, a cross-platform clinical app (Flutter/Dart/Firestore) for longitudinal tracking of developmental trends across neurodevelopmental studies
- Designed GUI tools and algorithmic pipelines for clinical data entry, eye-tracking QA, and behavioral assessments, supporting downstream statistical modeling workflows

•Software Development Intern

NASA Langley Research Center

Jun 2023 – Sep 2023

Hampton, VA

- Developed end-to-end data pipelines in Python to standardize and migrate legacy atmospheric science datasets to the NASA TOLNet repository, ensuring format compliance across multiple research instruments and sites
- Built automated data quality analytics to flag anomalies and inconsistencies in historical lidar datasets; delivered summary reports and documentation to cross-functional research teams

PROJECTS

•Bayesian Temporal Signal Extraction & Uncertainty Quantification

2025

R, RStan, RShiny, Variational Inference, LC-MS Mass Spectrometry

- Engineered a Bayesian deconvolution pipeline for sparse high-dimensional sensor arrays; designed automated QC using posterior uncertainty estimates to separate latent signals from stochastic noise and artifacts

•Audio Texture Synthesis via Convolutional Neural Networks

2023

Python, PyTorch, Generative AI, Signal Processing

- Adapted 2D computer vision style-transfer algorithms for 1D temporal audio data; built a CNN-based generative model from scratch using PyTorch to synthesize realistic audio textures via Gram matrix optimization

•Meteor Instrument – Embedded Radio Astronomy & Meteorology System

Ongoing

Python, Raspberry Pi, C++, Arduino, Embedded Systems, Signal Processing

- Developing a standalone embedded instrument on Raspberry Pi and Arduino display hardware for real-time radio astronomy and meteorology data capture, processing, and visualization

TECHNICAL SKILLS

Languages: Python, R, MATLAB, Julia, SQL, C++, JavaScript, Dart

Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, RStan, RShiny, Spark, Firestore, Git